

BUCHEL, Germany

WMO: 106130

Lat: 50.1747N

Lon: 7.0594E

Elev: 1594

StdP: 13.87

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.3	18.7	10.0	9.7	19.4	13.9	11.8	21.8	26.2	41.8	23.6	40.4	8.3	60	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	82.7	65.0	79.0	63.5	75.6	62.1	66.7	78.6	65.1	75.9	63.5	73.2	7.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	90.5	70.0	61.1	85.2	68.6	59.5	80.5	67.3	32.1	78.8	30.8	75.5	29.7	73.0	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	17.9	15.8	11.1	88.6	5.4	3.7	7.3	91.3	4.1	93.5	1.1	95.5	-2.8	98.2
			10.8	70.0	5.1	1.7	7.2	71.2	4.2	72.1	1.3	73.1	-2.3	74.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	32.4	34.1	40.1	47.2	53.8	59.7	63.5	62.6	55.9	48.4	39.9	34.1	(6)
	DBStd	12.99	7.45	7.36	7.22	7.58	7.09	6.91	6.84	6.57	5.96	6.46	6.53	7.28	(7)
	HDD50	2412	547	445	315	142	40	4	0	0	11	108	307	493	(8)
	HDD65	6470	1012	865	773	535	352	188	114	125	281	514	753	958	(9)
	CDD50	1579	0	0	7	58	159	294	418	392	188	59	4	0	(10)
	CDD65	161	0	0	0	0	6	28	67	52	8	0	0	0	(11)
	CDH74	1112	0	0	0	2	37	191	464	378	40	0	0	0	(12)
	CDH80	254	0	0	0	0	3	33	110	104	4	0	0	0	(13)
(14) Wind	WSAvg	7.9	8.9	8.9	8.8	7.9	7.6	7.2	7.2	6.9	7.2	7.9	8.0	8.8	(14)
(15) Precipitation	PrecAvg	28.5	2.4	2.1	2.0	1.8	2.5	2.7	2.6	2.4	2.3	2.4	2.4	2.7	(15)
	PrecMax	37.1	7.2	5.9	5.2	3.6	4.7	4.9	7.0	5.3	5.4	6.2	4.6	7.8	(16)
	PrecMin	21.8	0.4	0.2	0.2	0.0	0.7	0.7	1.1	0.5	0.6	0.8	0.2	0.5	(17)
	PrecStd	4.0	1.3	1.3	1.2	0.9	1.1	1.2	1.3	1.3	1.3	1.2	1.0	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	53.5	62.9	72.2	78.5	84.3	88.0	89.3	79.1	68.8	57.2	50.4	(19)
		MCWB	47.6	44.4	49.7	55.7	62.4	65.8	65.9	66.7	62.9	59.0	53.2	48.4	(20)
	2%	DB	47.7	48.5	57.6	67.8	73.2	79.1	83.0	82.4	73.5	63.7	53.4	48.5	(21)
		MCWB	45.9	43.7	47.2	53.3	59.6	64.2	64.8	64.9	61.4	57.0	50.3	47.0	(22)
	5%	DB	44.8	46.2	53.6	63.0	69.5	75.3	79.0	77.9	69.3	60.0	51.0	46.4	(23)
		MCWB	43.4	43.4	46.1	51.4	56.8	62.2	63.6	62.8	59.6	55.5	48.7	45.2	(24)
	10%	DB	42.3	44.0	50.1	59.2	65.8	71.8	75.4	73.8	65.8	57.1	48.5	44.1	(25)
		MCWB	41.0	41.8	44.6	49.7	55.4	60.7	62.0	61.7	58.1	53.9	46.9	43.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	47.2	51.4	57.3	64.5	69.1	68.7	68.4	64.9	60.3	53.9	49.4	(27)
		MCDB	49.5	49.5	58.3	68.3	75.8	79.6	82.4	82.0	74.2	65.7	55.9	50.0	(28)
	2%	WB	45.9	45.2	49.2	54.6	61.2	65.6	66.5	66.3	62.6	58.0	51.0	47.2	(29)
		MCDB	46.9	47.3	54.9	65.3	70.6	76.0	78.7	79.1	71.4	62.0	52.9	48.2	(30)
	5%	WB	43.3	43.5	47.3	52.1	58.5	63.6	65.0	64.5	60.6	56.1	49.0	45.1	(31)
		MCDB	44.6	45.4	52.1	60.7	67.2	73.4	75.9	74.8	67.3	59.4	50.6	46.1	(32)
	10%	WB	40.8	41.7	45.4	50.4	56.2	61.6	63.4	62.8	58.7	54.1	47.0	42.8	(33)
		MCDB	41.8	43.8	49.5	58.2	63.9	69.8	73.0	71.5	64.3	56.9	48.4	43.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.8	11.7	14.3	14.6	15.5	15.4	15.2	13.1	10.5	7.6	6.9	(35)
		MCDBR	9.5	11.9	16.7	19.0	19.0	20.1	20.1	20.3	17.7	14.0	9.9	7.8	(36)
	5% WB	MCWBR	8.5	8.5	9.6	9.8	9.8	10.2	8.3	8.8	8.9	8.7	7.6	7.2	(37)
		MCWBR	8.8	10.0	14.1	17.1	17.4	18.5	18.1	17.5	15.7	12.6	9.2	7.5	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.301	0.326	0.357	0.386	0.392	0.395	0.395	0.390	0.363	0.347	0.322	0.299	(40)
		taud	2.441	2.390	2.322	2.279	2.289	2.299	2.291	2.333	2.392	2.431	2.439	2.429	(41)
	Ebn at Noon	Ebn at Noon	237	257	267	272	274	274	272	267	263	246	224	219	(42)
		Edh at Noon	19	26	33	38	40	40	39	36	30	25	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	246	470	835	1323	1557	1723	1631	1400	1044	605	279	189	(44)
	RadStd	RadStd	38	89	128	160	152	160	183	140	129	60	35	28	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +0.83	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) -293	(l) +119	(m) N/A	(46)
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page